

Substitution Cipher **vs** Transposition Cipher

A comparative study of two foundational classical encryption methods — how they work, how they differ, and why they still matter in modern cryptography.



Confidentiality



Key-Based



Reversible



Foundational

Introduction

Understanding the building blocks of classical cryptography



What is Cryptography?

The science of transforming readable data (plaintext) into unreadable form (ciphertext) to keep information secure.



Two Core Families

Classical ciphers fall into two main categories: substitution ciphers and transposition ciphers.



Substitution Cipher

Replaces each letter of the message with another letter, number, or symbol using a fixed rule.



Transposition Cipher

Keeps the original letters but changes the order in which they appear.

Together, these two techniques form the foundation on which modern encryption algorithms like DES and AES are built.

Working Principle

How each cipher fundamentally transforms a message



Substitution Cipher

- ✦ Each letter is replaced by another letter, number, or symbol
- ✦ The position of the letter in the message stays exactly the same
- ✦ Uses a fixed key or rule, e.g. a shift value or a lookup table
- ✦ Changes the identity of characters, not their order



Transposition Cipher

- ✦ Letters remain exactly the same, only their positions change
- ✦ The message is rearranged using a grid, pattern, or key
- ✦ Original letters are never replaced, only reordered
- ✦ Changes the position of characters, not their identity

Encryption Process

Step-by-step: turning plaintext into ciphertext

Substitution Encryption

- 1 Choose a key (e.g. a shift number or a substitution table)
- 2 Take each letter of the plaintext one at a time
- 3 Replace it with the corresponding letter using the key
- 4 Combine all replaced letters to form the ciphertext

Transposition Encryption

- 1 Choose a key (e.g. number of columns and their read order)
- 2 Write the plaintext into a grid, row by row
- 3 Read the letters column by column in the key's order
- 4 Combine them together to form the ciphertext

Decryption Process

Reversing the transformation to recover the original message

Substitution Decryption

- 1 Use the exact same key that was used for encryption
- 2 Take each ciphertext letter one at a time
- 3 Replace it with the original letter (reverse the shift/table)
- 4 Combine the letters to recover the plaintext

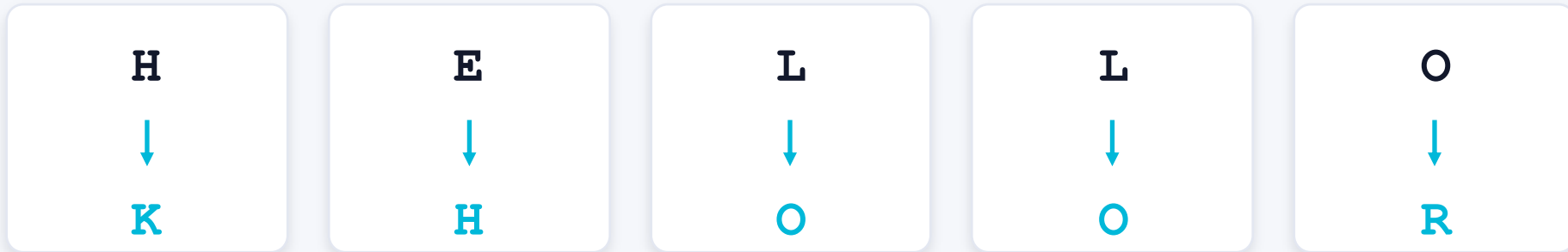
Transposition Decryption

- 1 Use the same key (column order) used for encryption
- 2 Rebuild the grid using the ciphertext and the key
- 3 Read the grid row by row instead of column by column
- 4 Combine the letters to recover the plaintext

Step-by-Step Example — Substitution

Caesar Cipher, using a shift key of 3

Plaintext: **HELLO** Key (shift): **3**



plaintext letter → shifted forward by 3 in the alphabet → ciphertext letter

Ciphertext: **KHOOR**

Step-by-Step Example — Transposition

Columnar Transposition, using a 4-column key order 2-4-1-3

Plaintext: **HELLOWORLDXX** (padded with X to fill the grid)

key order

2	4	1	3
H	E	L	L
O	W	O	R
L	D	X	X

Write the plaintext into the grid row by row

Read columns in key order → 2, 4, 1, 3

Col 2	Col 4	Col 1	Col 3
EWD	LRX	HOL	LOX

Ciphertext: **EWDLRXHOLLOX**

Advantages

What each cipher does well



Substitution Cipher

- ◆ Simple and fast to implement
- ◆ Easy to understand — a great intro to encryption concepts
- ◆ Can be made far more complex with multiple substitution alphabets
- ◆ Forms the basis of modern S-box design, e.g. AES SubBytes



Transposition Cipher

- ◆ Preserves the original letter frequency, hiding some patterns
- ◆ Uses simple mathematical operations — fast to compute
- ◆ Flexible: key size can grow with grid dimensions
- ◆ Works very well when combined with substitution

Disadvantages

Where each cipher falls short



Substitution Cipher

- ◆ Vulnerable to frequency analysis — common letters give it away
- ◆ Very small key space in simple form (Caesar has only 25 keys)
- ◆ Repeated letters and words create detectable patterns
- ◆ Alone, it is easily broken using statistical methods



Transposition Cipher

- ◆ Letter frequency of the plaintext is unchanged — a giveaway clue
- ◆ Vulnerable to anagramming and brute-force rearrangement
- ◆ Weak on its own against a skilled cryptanalyst
- ◆ Larger keys need bigger grids, adding real complexity

Applications

Where these classical techniques are used, then and now



Historical Military Use

The Caesar cipher protected military and political messages in ancient Rome.



Vigenère Cipher

Substitution logic scales up into the polyalphabetic Vigenère cipher.



AES S-Box (SubBytes)

Modern AES encryption uses substitution logic in its SubBytes step.



WWI / WWII Communication

Rail fence and columnar transposition secured wartime communication.

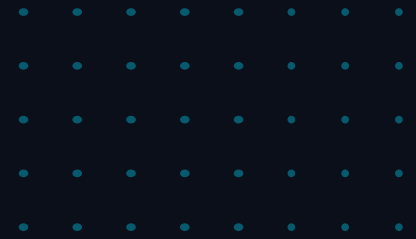


DES / AES Permutation

Transposition principles power the permutation steps in DES and AES.

Security Analysis

Comparing the strength of both techniques



	Substitution	Transposition
What changes	Character identity (position stays fixed)	Character position (identity stays fixed)
Main weakness	Frequency analysis of letters	Anagramming & pattern rearrangement
Key space (simple form)	Small — e.g. 25 shifts for Caesar	Grows with grid size, but still limited
Alone, is it secure?	No — broken by statistics	No — broken by rearrangement

Combining both principles — confusion (substitution) and diffusion (transposition) — is what makes modern ciphers like DES and AES strong.

Demonstration — Manual Calculation

Both techniques worked out by hand, side by side

Substitution — Shift Wheel Logic

A B C D E F G H I J K L M ...

shift each letter +3 places →

D E F G H I J K L M N O P ...

H is the 8th letter → shift to the 11th letter → **K**

Repeat for every letter of HELLO:

H→K E→H L→O L→O O→R

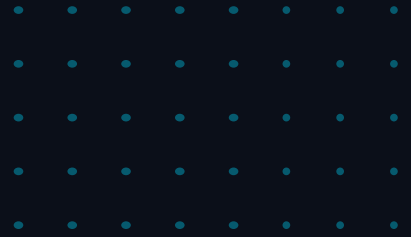
Transposition — Grid Read-Off Logic

1. Count letters, choose column count → **4 columns**
2. Pad the message so it fills the grid evenly
3. Write plaintext into the grid, row by row
4. Number the columns using the key → **2-4-1-3**
5. Read column 2, then 4, then 1, then 3
6. Concatenate to get the ciphertext:

EWDLRXHOLLOX

Conclusion

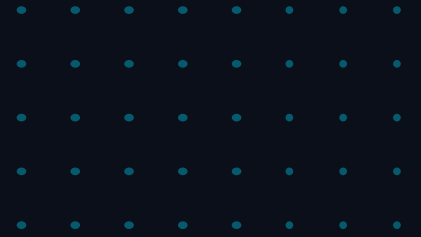
Two techniques, one shared foundation



- ✦ Substitution and transposition are the two fundamental building blocks of classical cryptography
- ✦ Substitution provides confusion; transposition provides diffusion — Shannon's core principles
- ✦ Neither technique is secure on its own by modern standards
- ✦ Together, they form the foundation of modern symmetric ciphers such as DES and AES
- ✦ Understanding these classical basics is the key to understanding all of modern encryption

Confusion (Substitution) + Diffusion (Transposition) = Strong Modern Encryption

THANK YOU



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